Abstract

Human Digital Twins (HDTs) — dynamic, data-driven virtual representations of individuals — have emerged as an influential concept across manufacturing, occupational health, affective computing, and, increasingly, healthcare. This review synthesizes twenty papers spanning two related literatures: (1) the foundational architecture, definitions, and cross-domain applications of HDTs, and (2) digital health and predictive-modeling approaches specifically targeting hypertensive disorders of pregnancy (HDP) and preeclampsia. We trace the evolution of HDT theory from conceptual disambiguation and networking architecture through to a recent scoping review showing that only a small fraction of published "digital twin" studies meet a strict definitional standard. We then examine how this theory intersects with real-world maternal-health tools, including mobile risk-triage platforms and home telemonitoring programs deployed in low-resource settings, alongside systematic evidence on their acceptability, effectiveness, and the current state of machine-learning preeclampsia prediction. We argue that maternal hypertension monitoring represents an instructive test case for HDT theory: existing tools already implement several architectural layers proposed in the HDT literature, yet fall short of full personalization, continuous updating, and validated predictive capability. We close by identifying concrete gaps — external validation of predictive models, integration of continuous physiological data streams, and context-sensitive design for low-resource settings — that must be addressed before genuine human digital twins for maternal health become clinically viable.

1. Introduction

The digital twin paradigm, first developed for monitoring and optimizing complex engineered systems such as aircraft and factories, has in the past decade been extended to represent human beings. This extension — the Human Digital Twin (HDT) — promises personalized, continuously updated virtual models capable of simulating, predicting, and informing decisions about an individual's health, behavior, or performance. Because HDTs sit at the intersection of sensing technology, data infrastructure, and domain-specific modeling, the literature describing them spans many application areas, from smart manufacturing and workplace safety to precision medicine and affective computing.

This review draws together twenty papers that collectively illustrate both the breadth of HDT theory and one increasingly important application area: monitoring and predicting hypertensive disorders of pregnancy (HDP), including preeclampsia. HDP affects a substantial share of pregnancies worldwide and remains a leading cause of maternal and perinatal morbidity and mortality, particularly in low- and middle-income countries (LMICs). The tools developed to address this problem — mobile risk-triage applications, home blood-pressure telemonitoring programs, and machine-learning prediction models — can be read as early, partial implementations of the more general HDT architecture described in the foundational literature. Examining them side by side allows us to ask a practical question: how close is the current state of digital maternal-health technology to the vision articulated by HDT theorists, and what would need to change to close that gap?

The remainder of this review proceeds as follows. Section 2 describes the scope and approach of the review. Section 3 examines conceptual foundations and definitional debates surrounding HDTs. Section 4 reviews proposed technical architectures. Section 5 surveys HDT and digital twin applications outside healthcare, which nonetheless inform healthcare-specific design. Section 6 turns to healthcare applications directly, including a critical scoping review of how well existing systems meet formal digital-twin criteria. Section 7 presents the maternal-health case study in depth. Section 8 synthesizes the two literatures, and Section 9 identifies gaps and future directions.

2. Scope and Approach

This review is a narrative synthesis rather than a formal systematic review. The twenty source papers were identified through targeted searches of academic databases and preprint repositories (arXiv, Springer, ScienceDirect, MDPI, PubMed/PMC, and JMIR) and were selected because they represent, in aggregate, the major strands of published work on (a) Human Digital Twin theory and architecture, (b) digital twin applications in adjacent domains such as manufacturing and occupational safety, and (c) digital health and predictive tools for hypertensive disorders of pregnancy. Several titles were represented more than once across earlier stages of this review process by minor variations in venue or wording; these have been consolidated so that each unique study is discussed once. Papers are grouped thematically below rather than strictly chronologically, since the goal is to trace conceptual and technical lineage rather than a strict publication timeline.

3. Conceptual Foundations of Human Digital Twins

3.1 Definitional Ambiguity and Disambiguation

The rapid spread of the HDT concept into new domains — especially healthcare — has produced substantial definitional drift. Gaffinet and colleagues conducted a systematic literature review across all application domains specifically to address this problem, finding broad consensus that the entity being twinned is always a human individual, but little agreement on how data flows between that individual and their digital counterpart. Their proposed resolution sorts systems into three tiers based on depth of data integration: human digital models (static, no automatic data exchange), human digital shadows (one-way, real-time data flow from the physical individual to the virtual model), and true human digital twins (bidirectional, continuously updated data exchange). They further introduce the notion of an "augmented HDT" for cases where a person is tightly coupled with a supporting technical system. This three-tier distinction becomes directly relevant later in this review, when we assess how many published "digital twin" healthcare systems actually qualify as such.

Lauer-Schmaltz and colleagues approached the same ambiguity from a design perspective, arguing that the field lacks not only a shared definition but also concrete guidance for building HDTs. Screening roughly seventy relevant studies, they propose a first cross-domain definition together with eleven key design considerations meant to guide future developers across healthcare, sports, and other fields — a more actionable complement to the taxonomic work of Gaffinet et al.

3.2 Broader Surveys of the Field

Lin and colleagues' widely cited survey screened 327 papers to characterize the overall shape of HDT research, proposing a generic system architecture organized around core functional blocks and reviewing state-of-the-art applications in healthcare, industry, and daily life. This survey, alongside the disambiguation and design-guidance papers above, forms the conceptual backbone that later domain-specific work — including the maternal-health tools discussed in Section 7 — implicitly or explicitly builds upon.

4. Technical Architecture of Human Digital Twins

4.1 Networking and Layered Architectures

Moving from definitions to implementation, Chen and colleagues' comprehensive survey on networking architecture for HDTs in personalized healthcare proposes a five-layer stack: data acquisition, data communication, computation, data management, and data analysis/decision-making. This layered framework is echoed in Okegbile and colleagues' complementary work on HDTs for personalized healthcare, which frames the HDT as consisting of a physical entity, a virtual model, and the connective infrastructure linking them, while emphasizing that the complexity of human physiology makes precise data extraction and modeling substantially harder than in industrial digital twin applications. Both papers explicitly flag remote monitoring, diagnosis, and personalized treatment as target use cases — directly foreshadowing the pregnancy-monitoring tools discussed later.

4.2 Human Body Modeling and Roadmaps

Tang and colleagues' Nature Reviews Electrical Engineering perspective focuses specifically on the human body digital twin, defined as a virtual representation of an individual's physiological state built from real-time sensor and medical-device data. Their proposed five-level roadmap runs from wearable sensing hardware through data collection, analysis, and clinical decision support, and explicitly addresses security, cost, and ethical considerations — concerns that resurface throughout the maternal-health literature reviewed in Section 7, particularly around data privacy and infrastructure limitations in low-resource settings.

He and colleagues extend this technical lineage by tracing the transition from earlier digital human modeling (DHM) tools — used for ergonomic and biomechanical analysis but limited in real-time responsiveness — to a proposed HDT framework combining a physical twin, a virtual twin integrating human modeling with AI, and the bidirectional link between them. Pan and colleagues' more recent contribution narrows this further to precision health, reviewing statistical and machine-learning approaches to HDT modeling, including anomaly detection and failure prediction, and discussing the ethical, technological, and regulatory barriers to deployment — many of which reappear in the ML preeclampsia literature discussed in Section 7.4.

5. Digital Twin Applications Beyond Healthcare

Although this review's ultimate focus is healthcare, several HDT applications in adjacent domains illustrate design patterns and challenges that recur throughout the healthcare literature.

5.1 Industry 5.0 and Manufacturing

Wang and colleagues situate the HDT concept within Industry 5.0's human-centric manufacturing paradigm, proposing a three-dimension model comprising the human entity, a virtual entity, and the interactive system linking them, and reviewing enabling technologies and applications in smart factories. This human-system coupling logic parallels the patient-clinician coupling seen in healthcare telemonitoring platforms, where a virtual representation of the patient's physiological state mediates the relationship between the patient and their care team.

5.2 Occupational Safety and Health

Park and colleagues survey human-focused digital twins for workplace safety, motivated by the observation that traditional occupational risk-assessment tools — checklists and expert evaluations — are manual and subjective. Their proposed digital-twin-based alternative would continuously collect and analyze worker data to flag hazards more objectively, though the authors note substantial integration challenges in combining the necessary hardware and software. This same tension between the promise of continuous, objective monitoring and the practical difficulty of integrating multimodal data recurs almost identically in the maternal telemonitoring literature.

5.3 Affective Computing and Digital Humans

Lu and Liu's work on affective digital twins addresses a different gap: existing digital humans in metaverse and virtual-reality contexts lack realistic emotional traits, limiting genuine human-machine interaction. Their proposed framework covers affective modeling, perception, encoding, and expression. While distant from clinical applications, this paper is a useful reminder that HDT research encompasses far more than physiological monitoring — it also concerns how virtual human representations behave and communicate, a dimension largely absent from current maternal-health tools but potentially relevant to future patient-facing conversational health assistants.

5.4 Datasets for LLM-Based Digital Twins

Toubia and colleagues' Twin-2K-500 dataset addresses a different bottleneck: the scarcity of large, public, individual-level ground-truth data needed to build and validate large-language-model-based digital twins of human behavior. Surveying 2,058 participants across 500 questions covering demographics, personality, cognition, and behavioral-economics tasks, the dataset is intended as a benchmark for LLM-based persona simulation. Although not healthcare-specific, it exemplifies a broader methodological requirement shared by clinical HDTs: robust, longitudinal, individual-level data is a precondition for any credible digital twin, a requirement that — as later sections show — is frequently unmet in maternal-health digital tools evaluated to date.

6. Human Digital Twins in Healthcare

6.1 General Frameworks and Conceptual Syntheses

Several papers attempt to organize the fragmented healthcare-HDT literature. The systematic meta-review of digital twins in healthcare synthesizes findings from twenty selected reviews (drawn from over a thousand candidate studies) using PRISMA methodology, extracting the functional, technological, and operational dimensions that characterize healthcare digital twins and how they relate hierarchically. This meta-review-of-reviews approach provides a useful conceptual map for the field, though — like much of the literature discussed below — it operates largely at the level of aspiration rather than validated clinical deployment.

6.2 Metaverse-Enabled Healthcare Digital Twins

Turab and Jamil examine digital twins operating within metaverse-style immersive environments, focused on personalized and precision medicine. They highlight potential applications in medical training and personalized care delivery, but also foreground technical, regulatory, and ethical challenges, including data privacy, standardization, and accessibility — concerns that reappear, in more concrete form, in the LMIC-focused maternal-health literature reviewed in Section 7.

6.3 Critical Appraisal: How Many Healthcare "Digital Twins" Really Qualify?

The most consequential paper for interpreting the rest of this literature is Tudor and colleagues' 2025 scoping review. Applying the National Academies of Sciences, Engineering, and Medicine (NASEM) definition — which requires that a genuine digital twin be personalized, dynamically updated, and predictive — the authors assessed 149 healthcare studies published between January 2017 and July 2024. Only eighteen studies, or roughly twelve percent, fully met all three NASEM criteria. Roughly a fifth of the remaining studies were better classified as digital shadows or general digital models, and virtual patient cohorts made up a further tenth. Notably, only two studies addressed verification, validation, and uncertainty quantification in any depth.

This finding substantially reframes how the rest of the literature should be read. Many systems described using digital-twin language — including several of the maternal-health tools discussed next — are more accurately characterized as digital health interventions, remote-monitoring platforms, or predictive-model deployments rather than digital twins in the strict engineering sense. This distinction is not merely semantic: it affects what claims can legitimately be made about personalization, continuous updating, and predictive reliability.

7. Case Study: Digital Health and HDT-Adjacent Approaches to Hypertensive Disorders of Pregnancy

7.1 Clinical Burden and Rationale

Hypertensive disorders of pregnancy, including preeclampsia, gestational hypertension, and chronic hypertension in pregnancy, affect up to roughly ten percent of pregnancies and are a leading cause of maternal and perinatal morbidity and mortality. The burden falls disproportionately on low- and middle-income countries, where maternal and perinatal mortality ratios remain substantially higher than in high-income settings, and where shortages of trained health workers and long travel distances to facilities delay case identification and treatment. This is precisely the context in which HDT-style continuous, remote, data-driven monitoring holds the greatest theoretical promise — and where, as shown below, practical implementation still lags considerably behind theory.

7.2 Mobile and Telemonitoring Tools

Two tools illustrate the practical, deployed end of this literature. PIERS on the Move, developed by Lim and colleagues, combines the miniPIERS clinical risk-prediction model with an integrated pulse-oximetry sensor (the "Phone Oximeter") into a low-cost mobile application designed to help frontline health workers in resource-limited settings triage hypertensive pregnant women. It was subsequently implemented at scale in the Community Level Interventions for Pre-eclampsia (CLIP) cluster-randomized trials across India, Pakistan, and Mozambique, where evaluations found good coverage and community acceptance of triage recommendations, but only limited apparent impact on maternal health outcomes overall — an early signal that acceptability and usability do not automatically translate into improved clinical endpoints.

Raabta, developed for use in Karachi, Pakistan, represents a more recent and more patient-centered evolution of the same idea. Rather than supporting provider-led triage, Raabta is a smartphone application paired with a Bluetooth blood-pressure device that enables pregnant women at high risk of preeclampsia to self-monitor blood pressure and symptoms at home, triggering automated alerts (for example, prompting a repeat reading for mild hypertension or an immediate emergency-department visit for severe hypertension) and feeding a clinician-facing dashboard for oversight between antenatal visits. A series of usability and implementation studies around Raabta identified recurring barriers: the need for clearer instructions and terminology, accessibility challenges for non-tech-savvy and illiterate Urdu-speaking users, and greater apparent feasibility of sustained adoption in private-sector than public-sector hospital settings, driven by differences in patient demographics, digital literacy, and hospital infrastructure.

Read against the HDT architecture described in Sections 3 and 4, both tools implement recognizable pieces of the proposed data-acquisition and communication layers, and Raabta in particular begins to approach the "dynamically updated" criterion from the NASEM definition through daily self-monitoring. Neither tool, however, incorporates the kind of individualized predictive modeling or bidirectional, multimodal data integration that would qualify as a full human digital twin under the stricter taxonomies discussed in Section 3.1 — both are more accurately described as structured digital health interventions or, at most, digital shadows.

7.3 Evidence Synthesis Across Digital Health Interventions

Zooming out from individual tools, a 2025 systematic review of eHealth interventions for hypertensive disorders of pregnancy synthesized 100 publications covering 61,539 participants. Its central finding was that while participants generally found such interventions feasible, safe, and acceptable, with a broadly positive patient experience, the underlying evidence was of low quality and insufficient to support routine clinical recommendation. There was only limited evidence that eHealth tools reduce healthcare utilization, improve follow-up and blood-pressure ascertainment, reduce hospital admissions, or provide clear economic benefit relative to usual care.

Complementary scoping reviews focused specifically on low- and middle-income countries reach a similar, cautiously optimistic-but-unproven conclusion. These reviews identify persistent structural barriers — poor infrastructure, weak integration between digital tools and existing hospital information systems, limited buy-in from obstetricians and gynecologists, and gaps in digital literacy among both patients and health workers — while also emphasizing that human-centered, culturally and linguistically adapted design (mirroring the specific usability findings from the Raabta studies) is essential for adoption in these settings.

7.4 Predictive Modeling: Machine Learning for Preeclampsia

The final piece of this case study concerns the predictive-modeling component that any future maternal-health HDT would need. A 2026 systematic review and meta-analysis of machine-learning prediction models for preeclampsia pooled 26 studies comprising 31 distinct models and found a high pooled area under the receiver operating characteristic curve of 0.91. However, this headline figure obscured extreme heterogeneity across studies (I-squared exceeding 99 percent) and a wide 95 percent prediction interval for sensitivity, ranging from 0.32 to 0.96 — meaning that in some plausible external settings, a model's true sensitivity could fall as low as thirty-two percent. Critically, only six of the twenty-six studies conducted external validation, and in those six, pooled sensitivity dropped to 0.68. The authors concluded that a high area under the curve in these models more often reflects favorable internal performance than genuine, externally transportable clinical utility. A related and slightly earlier systematic review of eleven studies covering over 116,000 pregnancies reached a broadly consistent conclusion: machine-learning approaches outperform traditional regression-based prediction models in principle, but methodological quality and external validation remain inconsistent across the field.

8. Synthesis: Bridging HDT Theory and Maternal Health Practice

Taken together, the two literatures reviewed here suggest a coherent, if still incomplete, developmental arc. The foundational HDT papers (Sections 3–4) establish what a genuine human digital twin should be: a personalized, continuously and bidirectionally updated, predictively capable virtual representation of an individual, built on a layered architecture spanning sensing, communication, computation, data management, and decision-making. The domain-application papers (Section 5) demonstrate that this architecture has already been productively adapted to manufacturing, occupational safety, and affective computing, each surfacing recurring implementation challenges — data integration difficulty, hardware-software heterogeneity, and the gap between conceptual promise and deployable systems.

The healthcare-specific literature (Section 6) shows that these same challenges are, if anything, more acute in clinical contexts, and the Tudor et al. scoping review provides a sobering empirical check: the overwhelming majority of published healthcare "digital twins" do not meet a rigorous definitional standard, most commonly falling short on dynamic updating, individual-level personalization, or validated predictive capability.

The maternal-health case study (Section 7) makes this gap concrete. PIERS on the Move and Raabta demonstrate real, deployed progress on the data-acquisition and communication layers of the HDT architecture, particularly in low-resource settings where the theoretical benefits of continuous, remote monitoring are greatest. Yet the broader evidence synthesis shows that acceptability and feasibility have outpaced proof of clinical benefit, and the machine-learning prediction literature shows that the predictive-modeling layer — arguably the most distinctive feature separating a true digital twin from a simple monitoring app — remains unreliable outside the specific datasets on which models were developed. In HDT terms, maternal hypertension monitoring today occupies something closer to the "digital shadow" category described by Gaffinet and colleagues: real-time, mostly one-directional data capture from patient to provider, without the fully personalized, validated, bidirectional predictive loop that would justify the label "digital twin" in the strict sense.

9. Gaps and Future Research Directions

Several concrete gaps emerge from this synthesis. First, external validation remains the single most consistent weakness across both the general HDT-in-healthcare literature and the preeclampsia-specific machine-learning literature; future predictive models for HDP should be evaluated on populations and settings distinct from their training data before any deployment claims are made. Second, most maternal-health digital tools reviewed here capture a narrow slice of physiological data (primarily blood pressure and symptom checklists) rather than the multimodal, continuously updated data streams envisioned in the human body digital twin roadmap; integrating additional signals — such as fetal monitoring data, laboratory biomarkers, or wearable-derived physiological measures — could move these tools closer to genuine digital-twin status. Third, the recurring usability findings from the Raabta studies, and the structural barriers identified in LMIC-focused scoping reviews, indicate that technical sophistication alone is insufficient; culturally and linguistically adapted, human-centered design remains a precondition for adoption in the very low-resource settings where these tools are most needed. Finally, the field would benefit from applying the disambiguation framework proposed by Gaffinet and colleagues consistently, so that future maternal-health publications distinguish clearly between digital models, digital shadows, and true digital twins rather than defaulting to the more marketable but often imprecise "digital twin" label.

10. Conclusion

Human Digital Twins represent an ambitious and theoretically well-developed paradigm for personalized, predictive, continuously updated virtual representations of individuals, with architectural foundations now established across manufacturing, occupational health, affective computing, and general healthcare. When this theory is tested against a concrete and consequential clinical problem — hypertensive disorders of pregnancy — the picture that emerges is one of genuine but partial progress. Deployed tools such as PIERS on the Move and Raabta demonstrate that key architectural layers of the HDT vision can be implemented in real, low-resource clinical settings, and machine-learning approaches show promising predictive performance on the datasets they were trained on. However, rigorous evidence syntheses reveal that clinical benefit remains unproven, external predictive validity remains weak, and — per the most direct empirical test available — only a small minority of healthcare systems calling themselves "digital twins" would meet a strict definitional standard. Closing this gap will require sustained attention to external validation, richer and more continuous data integration, and context-sensitive design, rather than further proliferation of the digital-twin label itself.

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